

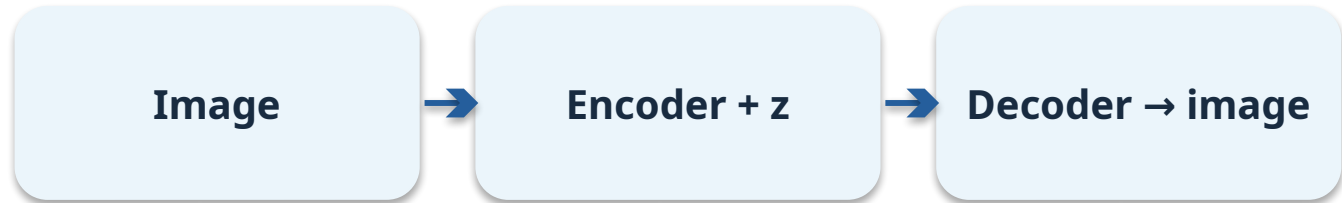
Week 02 · VAE and GAN model ideas

OPEN / WEEK 02

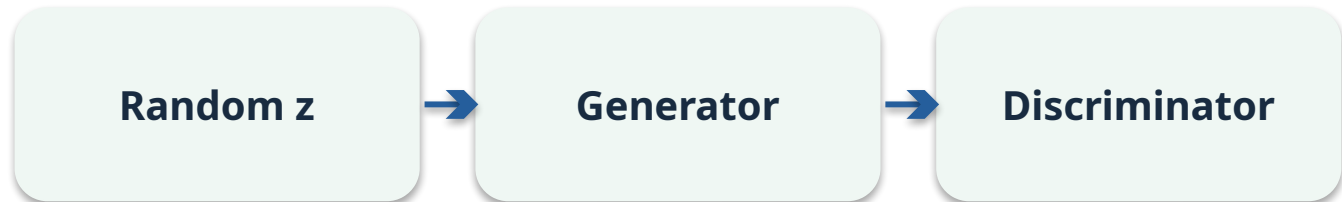
The icon studio

A designer wants varied icons that resemble a style without copying one example. How could two training ideas approach this?

VAE: reconstruct an example



GAN: improve a proposed example



Feedback updates both networks

First vote: what would you do?

QUESTION / WEEK 02

A designer wants varied icons that resemble a style without copying one example. How could two training ideas approach this?

Choose an initial answer and name one piece of evidence you need.

VOTE FIRST • GIVE A REASON

A Accept the first plausible result

B Check the source or measurement

Which source or test could settle this?

The journey through today's case

OPEN / WEEK 02

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.

1

VAE

2

Latent space

3

GAN

4

Training curves

1. Predict · VAE

QUESTION / WEEK 02

Which arrow could lose a fine detail?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Compression into z can discard information; compare original and reconstruction.

B A VAE stores and copies exact training icons.

Which source or test could settle this?

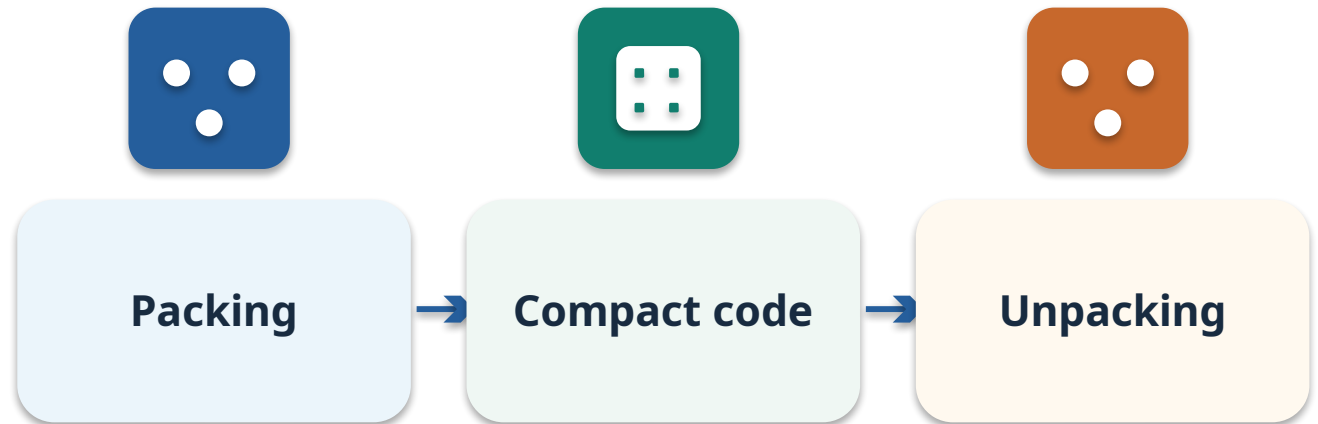
VAE · a familiar picture

ANALOGY / WEEK 02

A travel suitcase packs the essential features and unpacks them at the destination.

What matches our case, and where does the analogy stop?

A travel suitcase packs the essential features and unpacks them at the destination.



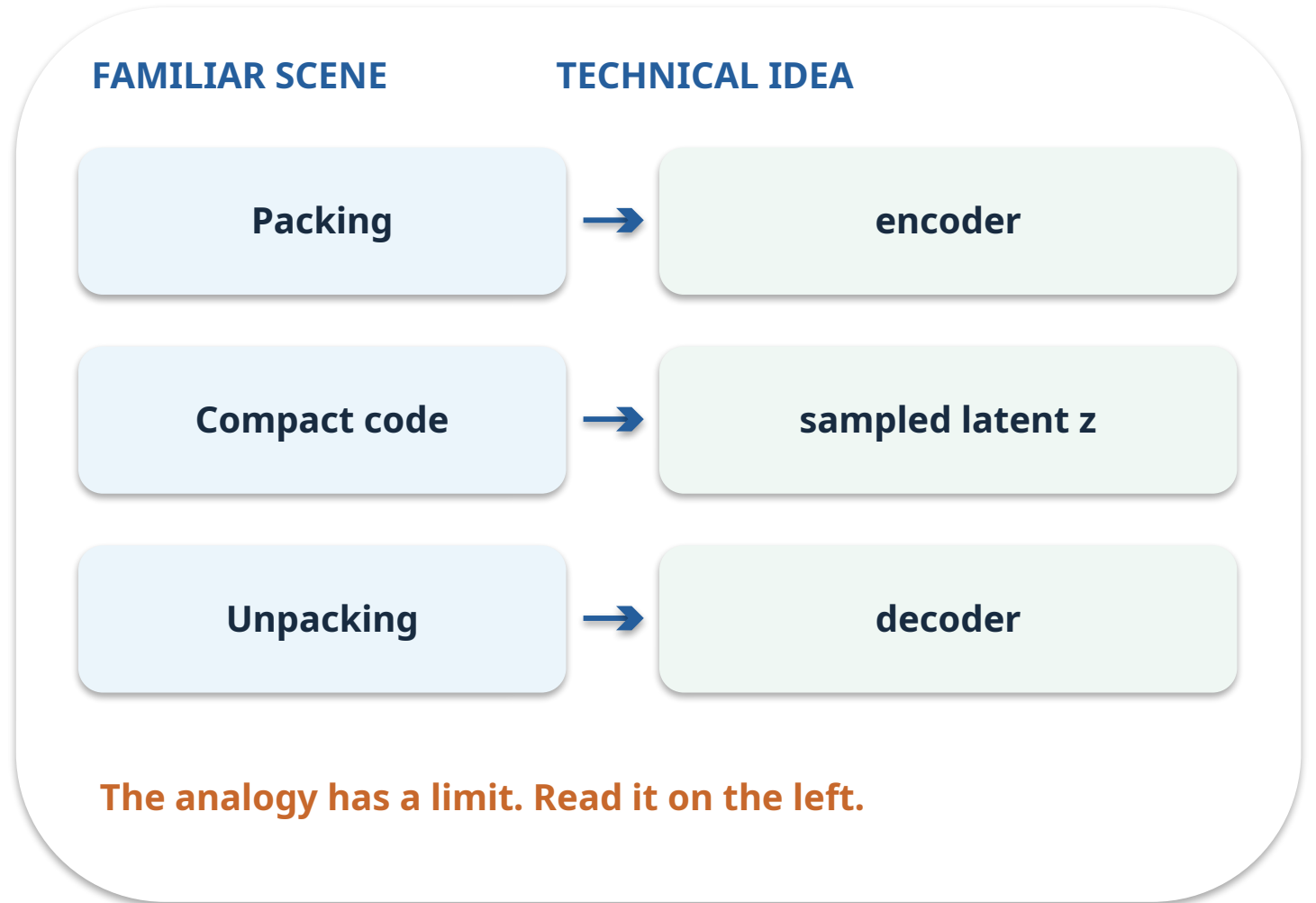
What matches the technical process?

VAE · map the parts

MAPPING / WEEK 02

Match each everyday part to the technical part.

Analogy limit: A VAE learns a distribution and reconstruction objective; a suitcase is not a probabilistic model.



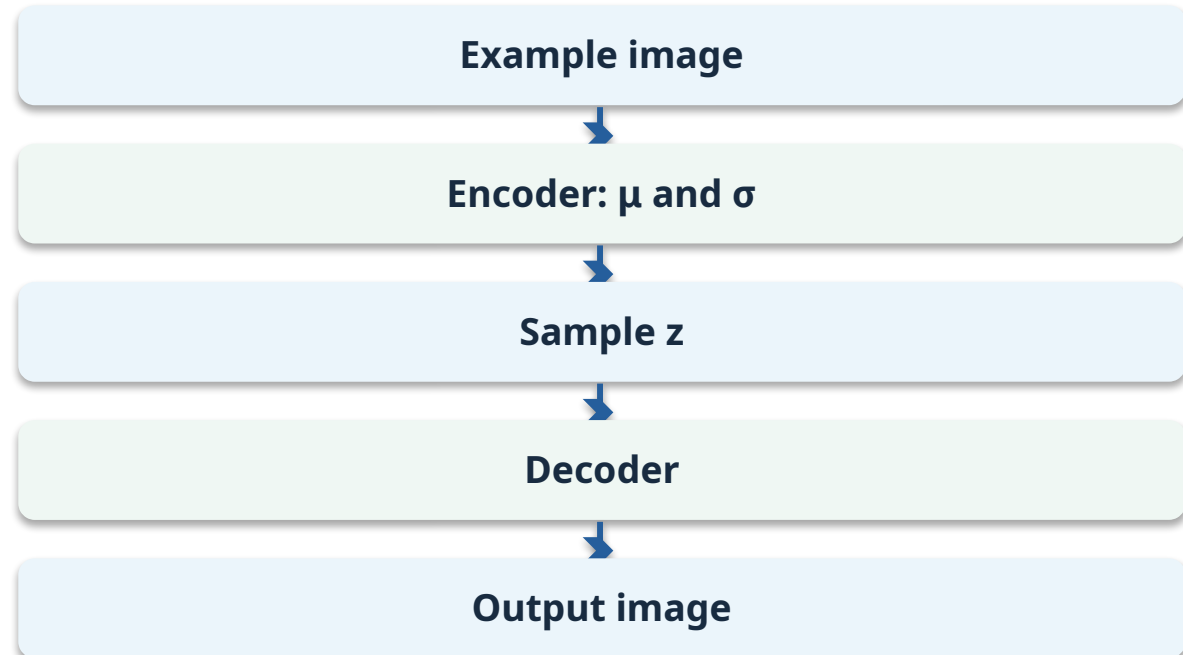
VAE · how it works

MECHANISM / WEEK 02

An encoder predicts the centre (μ) and spread (σ) of a small latent distribution. We sample a code z from it. A decoder uses z to reconstruct the image.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



VAE · work one case

WORKED / WEEK 02

A 2-value icon [0.2, 0.8] enters an encoder; a sampled z is decoded into a reconstructed icon.

Point to the evidence before giving your conclusion.

GIVEN

A 2-value icon [0.2, 0.8] enters an encoder; a sampled z is decoded into a reconstructed icon.

QUESTION

Which arrow could lose a fine detail?

SUPPORTED ANSWER

Compression into z can discard information; compare original and reconstruction.

VAE • answer and explain

REVEAL / WEEK 02

Compression into z can discard information; compare original and reconstruction.

Tempting mistake: A VAE stores and copies exact training icons.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A VAE stores and copies exact training icons.

SUPPORTED EXPLANATION

Compression into z can discard information; compare original and reconstruction.

A VAE learns a distribution and reconstruction objective; a suitcase is not a probabilistic model.

2. Predict · Latent space

QUESTION / WEEK 02

Does a smoother output prove realism?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A No. Inspect examples and use a suitable quality test.

B A latent coordinate is a pixel location.

Which source or test could settle this?

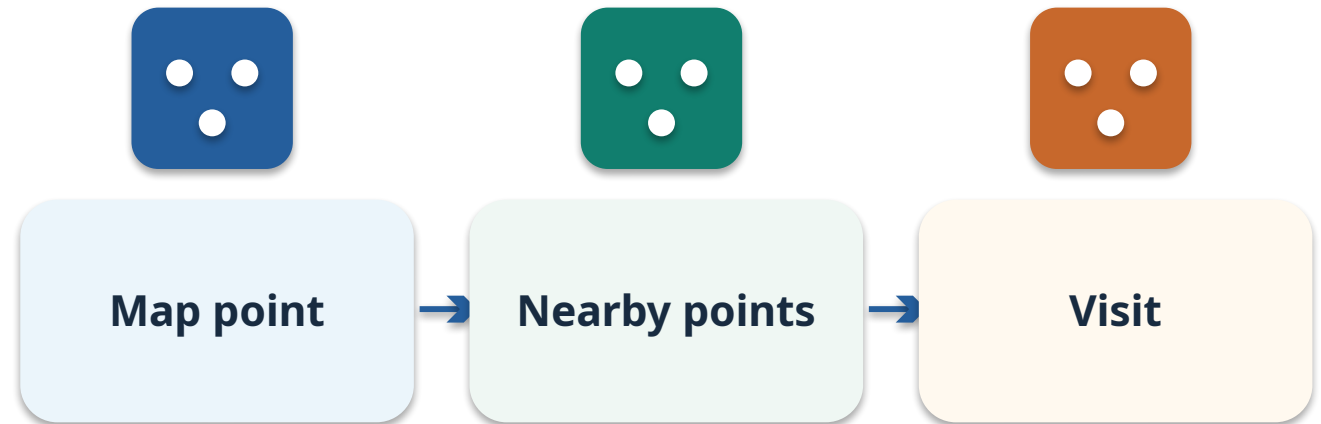
Latent space · a familiar picture

ANALOGY / WEEK 02

A map places similar destinations near one another.

What matches our case, and where does the analogy stop?

A map places similar destinations near one another.



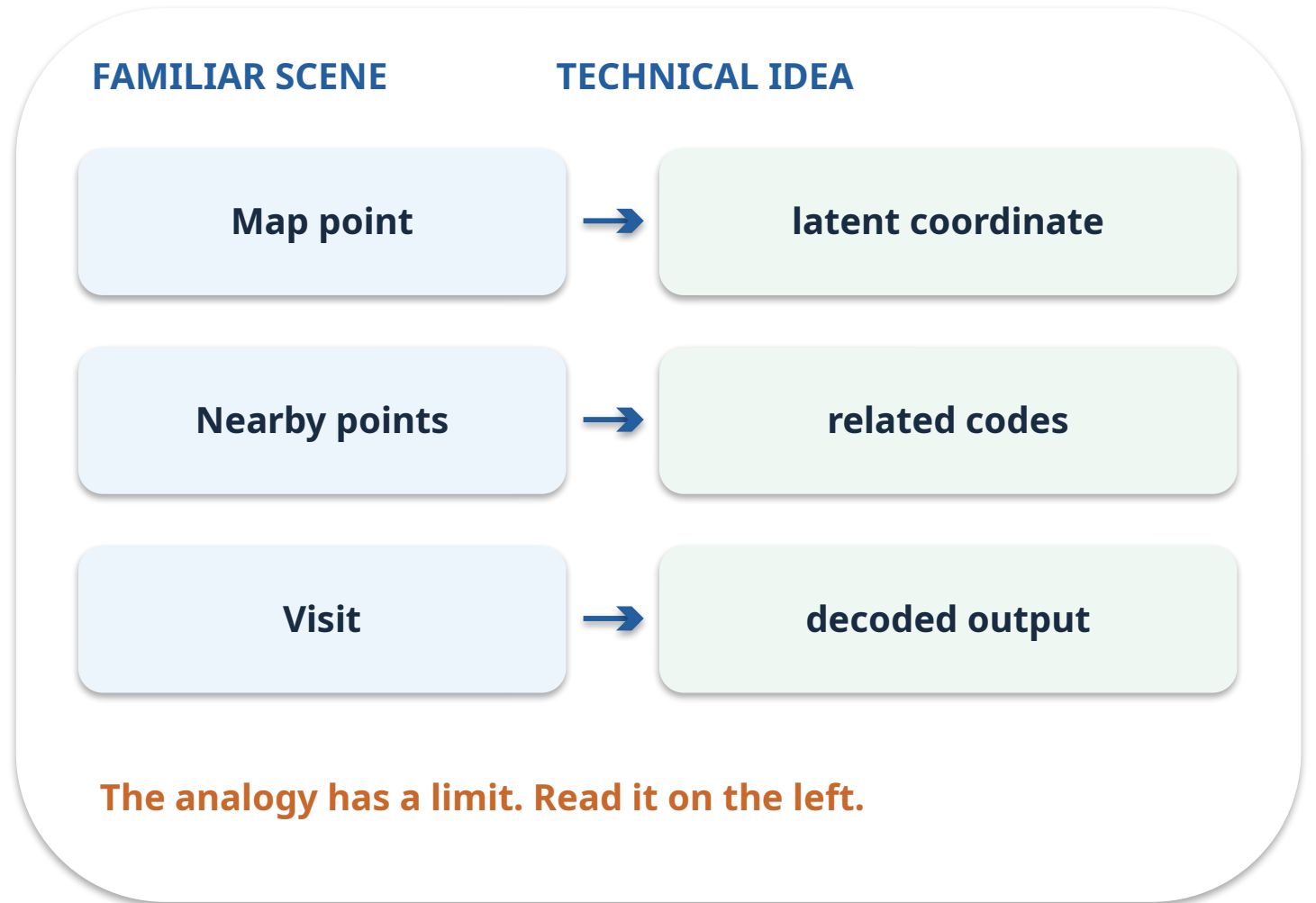
What matches the technical process?

Latent space · map the parts

MAPPING / WEEK 02

Match each everyday part to the technical part.

Analogy limit: Closeness in a toy plot does not guarantee human-perceived similarity.



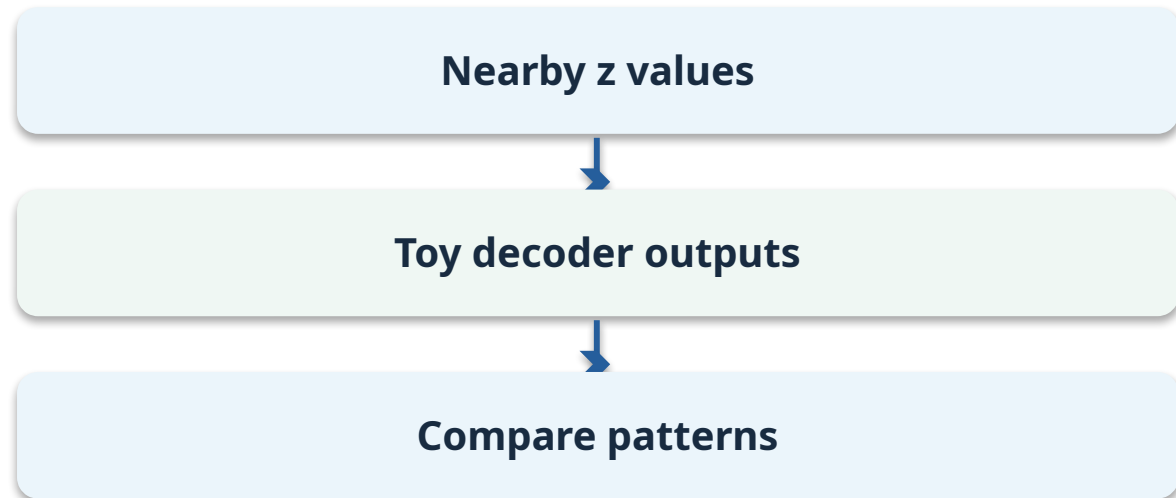
Latent space · how it works

MECHANISM / WEEK 02

Nearby latent values often produce related outputs, though this is learned rather than guaranteed.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Latent space · work one case

WORKED / WEEK 02

Change toy z from 0.3 to 0.7;
compare the output pattern
while keeping the decoder fixed.

Point to the evidence before
giving your conclusion.

GIVEN

Change toy z from 0.3 to 0.7; compare the output pattern while keeping the decoder fixed.

QUESTION

Does a smoother output prove realism?

SUPPORTED ANSWER

No. Inspect examples and use a suitable quality test.

Latent space · answer and explain

REVEAL / WEEK 02

No. Inspect examples and use a suitable quality test.

Tempting mistake: A latent coordinate is a pixel location.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A latent coordinate is a pixel location.

SUPPORTED EXPLANATION

No. Inspect examples and use a suitable quality test.

Closeness in a toy plot does not guarantee human-perceived similarity.

3. Predict · GAN

QUESTION / WEEK 02

Which part produces the new image?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A The generator produces it; the discriminator provides a training signal.

B The discriminator paints the final image.

Which source or test could settle this?

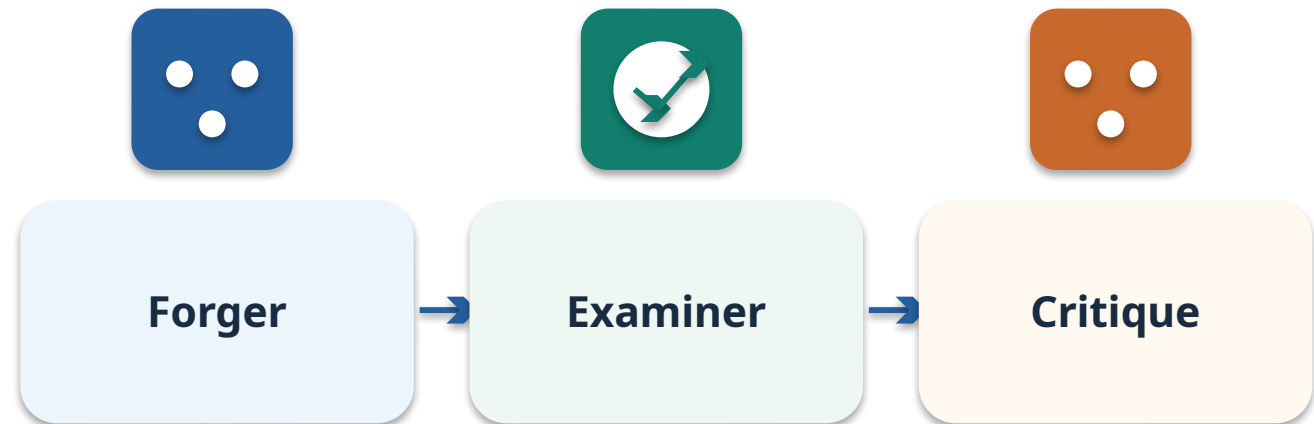
GAN · a familiar picture

ANALOGY / WEEK 02

A forger and examiner improve through repeated feedback.

What matches our case, and where does the analogy stop?

A forger and examiner improve through repeated feedback.



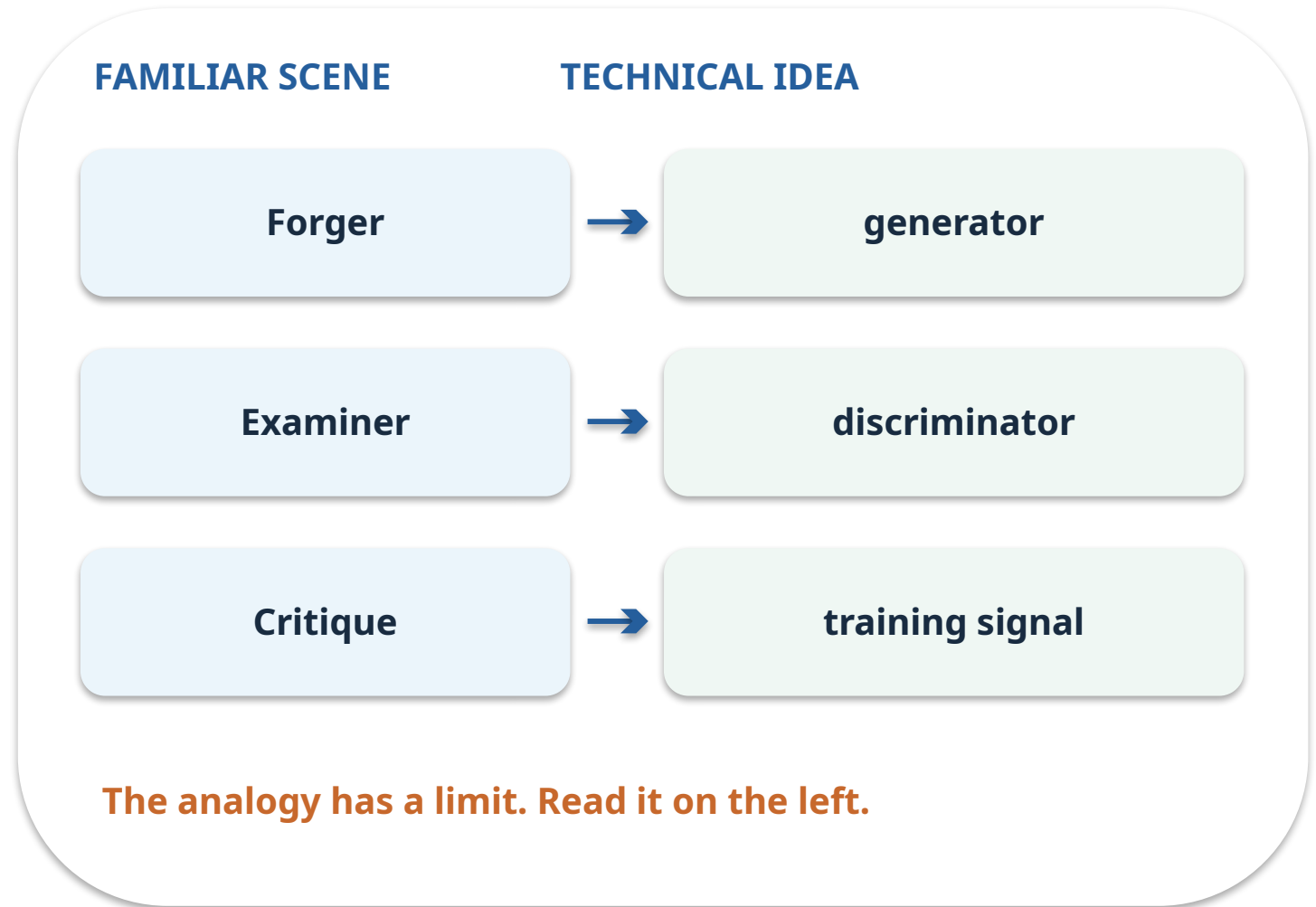
What matches the technical process?

GAN · map the parts

MAPPING / WEEK 02

Match each everyday part to the technical part.

Analogy limit: The analogy can imply a human examiner; the discriminator is a learned network and may be fooled.



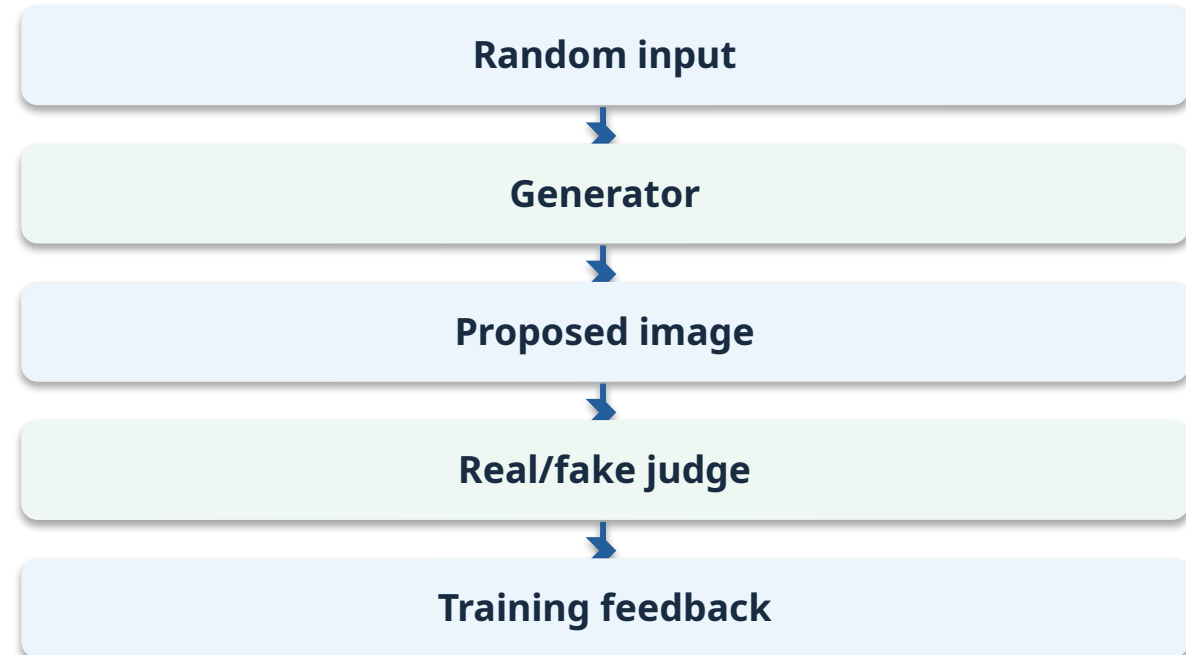
GAN · how it works

MECHANISM / WEEK 02

A generator proposes samples. A discriminator learns to distinguish generated and real samples; both change during training.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



GAN · work one case

WORKED / WEEK 02

A generator proposes an icon; the discriminator scores real versus generated examples; both update.

Point to the evidence before giving your conclusion.

GIVEN

A generator proposes an icon; the discriminator scores real versus generated examples; both update.

QUESTION

Which part produces the new image?

SUPPORTED ANSWER

The generator produces it; the discriminator provides a training signal.

GAN · answer and explain

REVEAL / WEEK 02

The generator produces it; the discriminator provides a training signal.

Tempting mistake: The discriminator paints the final image.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The discriminator paints the final image.

SUPPORTED EXPLANATION

The generator produces it; the discriminator provides a training signal.

The analogy can imply a human examiner; the discriminator is a learned network and may be fooled.

4. Predict · Training curves

QUESTION / WEEK 02

Why is the final training number insufficient?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Validation worsened; quality and variety require separate inspection.

B Lower training loss always proves better new icons.

Which source or test could settle this?

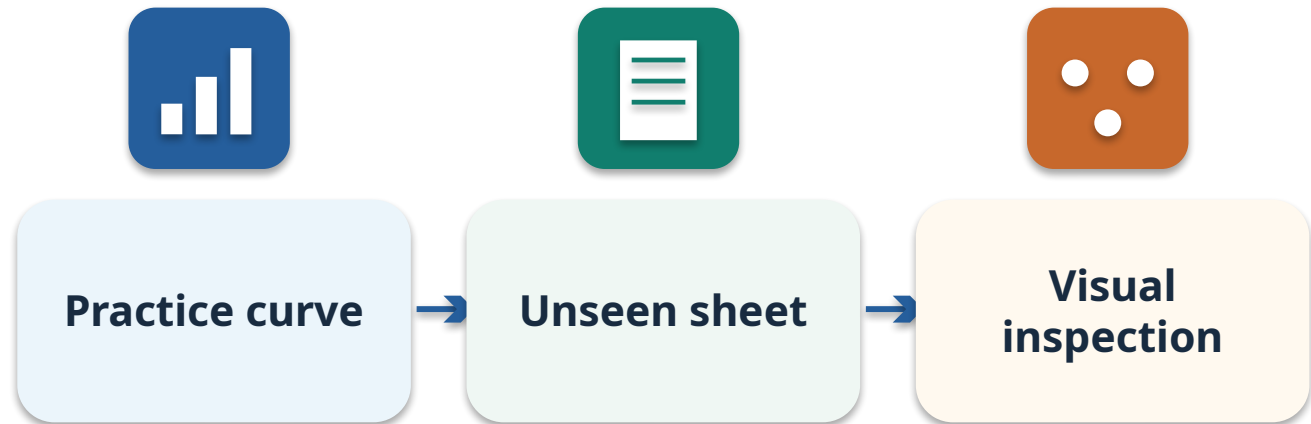
Training curves · a familiar picture

ANALOGY / WEEK 02

A student gets better on practice sheets while worse on unfamiliar exam sheets.

What matches our case, and where does the analogy stop?

A student gets better on practice sheets while worse on unfamiliar exam sheets.



What matches the technical process?

Training curves · map the parts

MAPPING / WEEK 02

Match each everyday part to the technical part.

Analogy limit: GAN losses are not simple universal quality scores; compare held-out samples too.

FAMILIAR SCENE

Practice curve

Unseen sheet

Visual inspection

TECHNICAL IDEA

training loss

validation set

quality check

The analogy has a limit. Read it on the left.

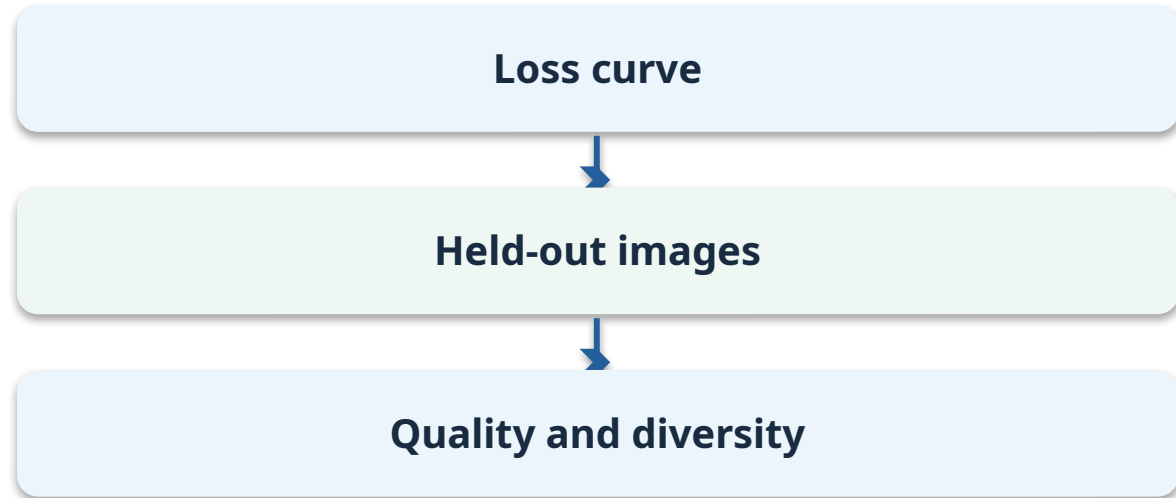
Training curves · how it works

MECHANISM / WEEK 02

A lower loss alone does not certify useful images; inspect quality, diversity and a held-out sample.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Training curves · work one case

WORKED / WEEK 02

Training error $0.30 \rightarrow 0.18 \rightarrow 0.12$;
validation $0.32 \rightarrow 0.20 \rightarrow 0.26$.
Inspect epoch 3.

Point to the evidence before
giving your conclusion.

GIVEN

**Training error $0.30 \rightarrow 0.18 \rightarrow 0.12$; validation $0.32 \rightarrow 0.20 \rightarrow 0.26$.
Inspect epoch 3.**

QUESTION

Why is the final training number insufficient?

SUPPORTED ANSWER

Validation worsened; quality and variety require separate inspection.

Training curves · answer and explain

REVEAL / WEEK 02

Validation worsened; quality and variety require separate inspection.

Tempting mistake: Lower training loss always proves better new icons.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Lower training loss always proves better new icons.

SUPPORTED EXPLANATION

Validation worsened; quality and variety require separate inspection.

GAN losses are not simple universal quality scores; compare held-out samples too.

Evidence trail: input → result

WORKED / WEEK 02

Latent coordinate z changes from 0.3 to 0.7.

Toy decoder output changes; inspect rather than assume improvement.

GIVEN

Input icon represented by [0.2, 0.8].

QUESTION

Latent coordinate z changes from 0.3 to 0.7.

SUPPORTED ANSWER

Toy decoder output changes; inspect rather than assume improvement.

Notebook: run the prepared example

PRACTICE / WEEK 02

Annotate model paths and explore a toy latent coordinate.

Read the original output before editing.

1

Trace an example through encoder, sample and decoder

2

Trace random input through generator and discriminator

3

Inspect supplied loss and quality figures

4

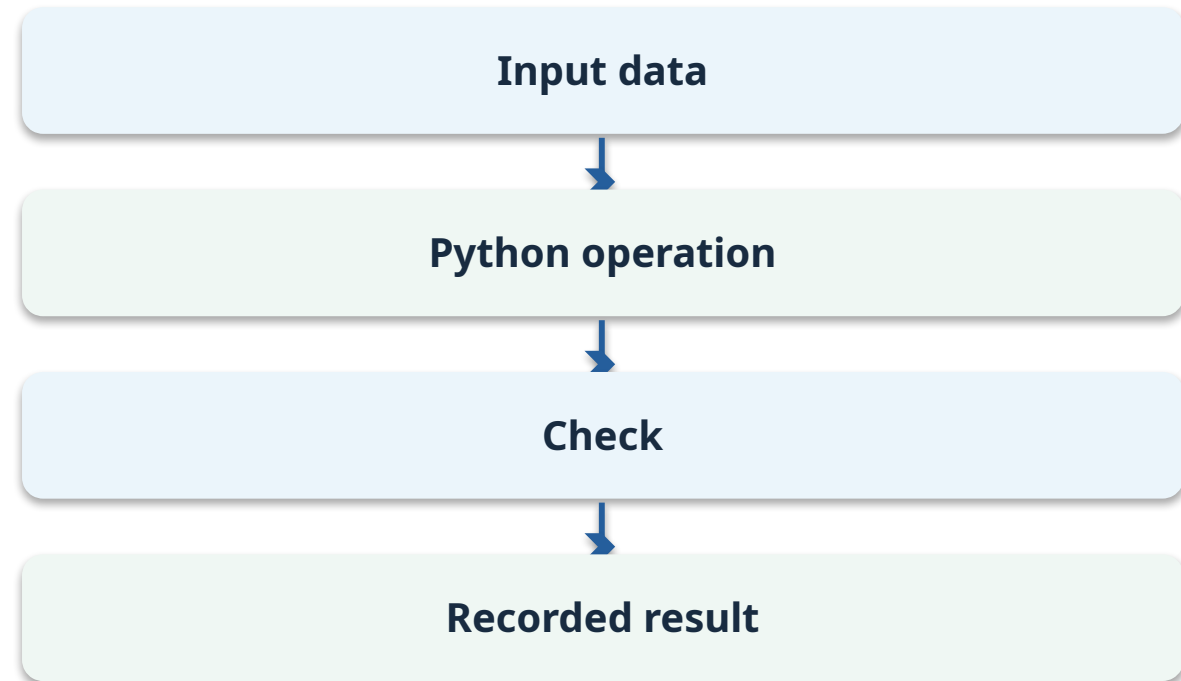
Change one latent coordinate in a toy visualization

Find the input and check in Python

PRACTICE / WEEK 02

Locate the supplied input, the transformation, the checking line and the printed result.

FOLLOW THE DATA THROUGH EACH ARROW



Predict one controlled change

PRACTICE / WEEK 02

Change z and compare output difference; explain that the toy cannot establish realism.

What should change in the result? What stays fixed?

Before: original case

After: one change

Keep all other inputs fixed.

Run, compare and record

PRACTICE / WEEK 02

Record the input, the one changed setting, the observed result and a reason for any difference.

Input	Setting	Result
Original	baseline	record
Changed	one edit	compare

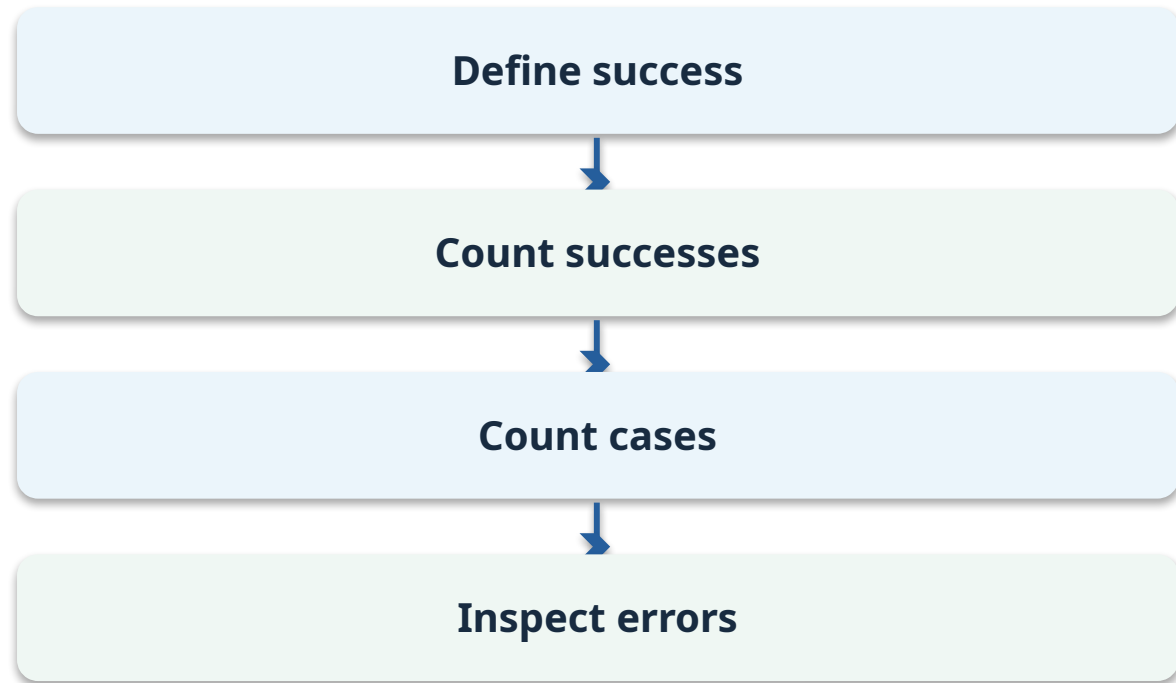
Measure without hiding the denominator

CHECK / WEEK 02

Mean absolute reconstruction error on supplied example pairs

Write the numerator, denominator and one failure case.

FOLLOW THE DATA THROUGH EACH ARROW



Error detective: find the layer

CHECK / WEEK 02

Plausible wrong claim: The toy decoder is a trained VAE or a trained GAN.

Correct it using the case evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The toy decoder is a trained VAE or a trained GAN.

SUPPORTED EXPLANATION

It is only a deterministic illustration of latent variation. Real models learn parameters from examples.

Name a test that would catch this failure.

Transfer to a new situation

CHECK / WEEK 02

The generated icon looks sharp but repeats one shape. Which additional check would reveal lack of diversity?

Explain which source or test would settle it.

VOTE FIRST • GIVE A REASON

A Use the visible evidence

B Flag missing or unsafe evidence

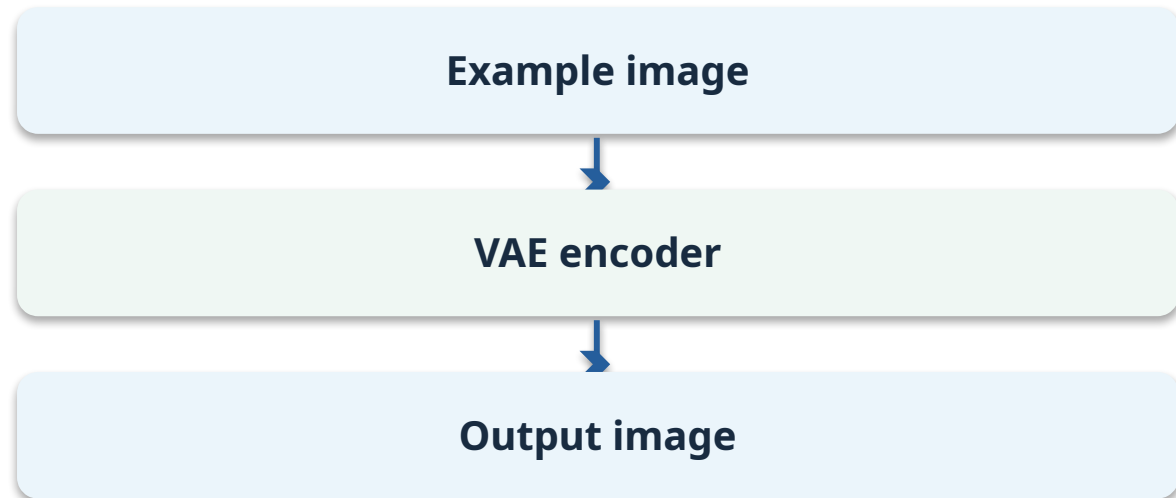
Which source or test could settle this?

Explain it to a partner

CHECK / WEEK 02

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

FOLLOW THE DATA THROUGH EACH ARROW



Exam rehearsal · explain and apply

CHECK / WEEK 02

Label input, encoder, latent sample and decoder in a VAE.

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

Exam rehearsal · explain and apply

CHECK / WEEK 02

Training error falls but validation error rises. What should you inspect?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

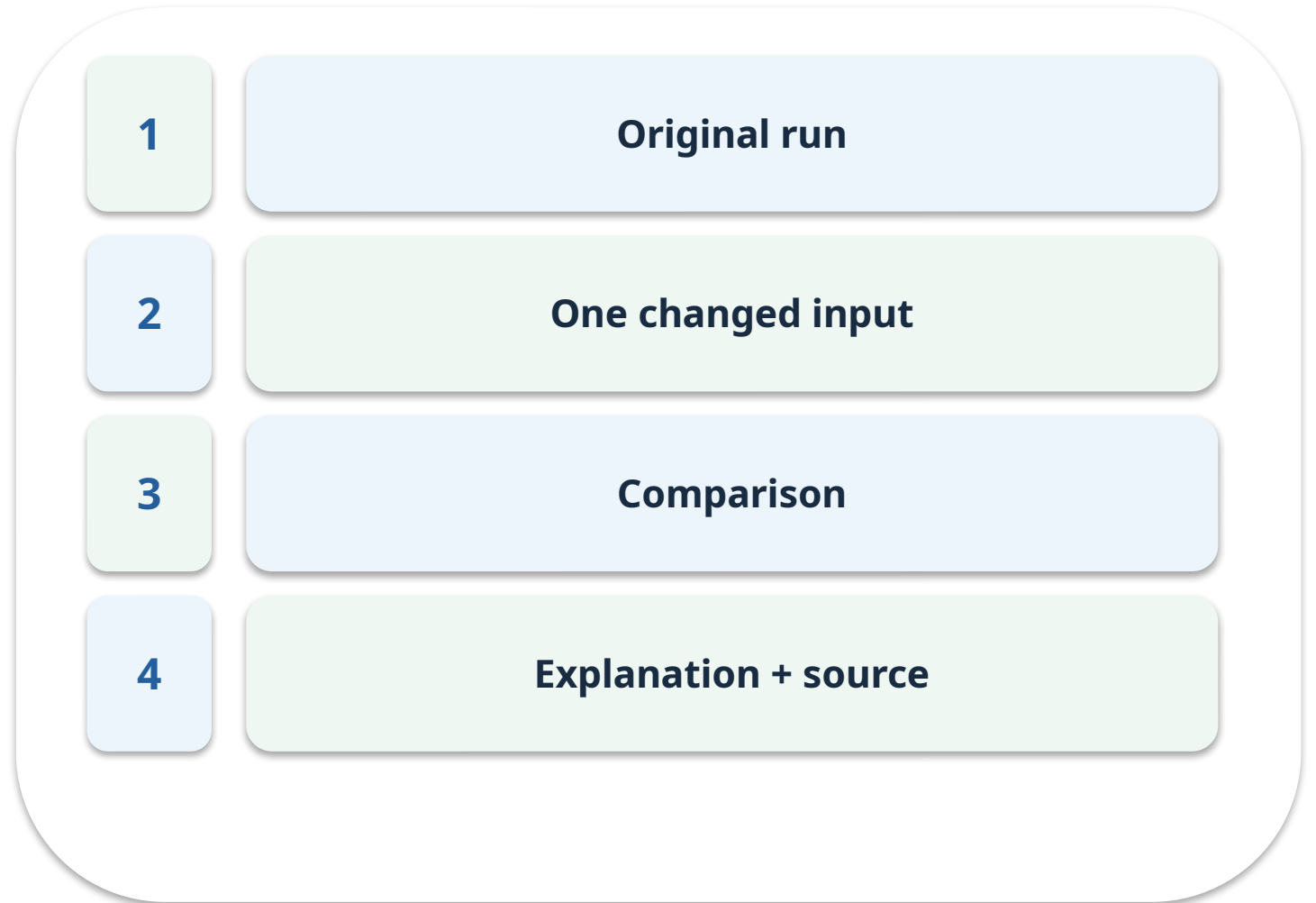
Compare with the Word guide

What to hand in

PRACTICE / WEEK 02

Two labelled model diagrams and a reconstruction comparison.

Include the one changed input and an error explanation.



Three ideas to recall next week

CHECK / WEEK 02

VAE: compress and reconstruct.
GAN: propose and critique.

Read the linked official sources
in the Word guide and redraw
one diagram from memory.

1

VAE

2

Latent space

3

GAN

4

Training curves